Vite Build for React

When building a new React project, it's important to choose the right tool for project setup and development. In the past, developers commonly used Create React App (CRA) to quickly scaffold React applications. However, Create React App is now deprecated, and Vite has become the preferred alternative for setting up modern React projects.

Introduction to Vite

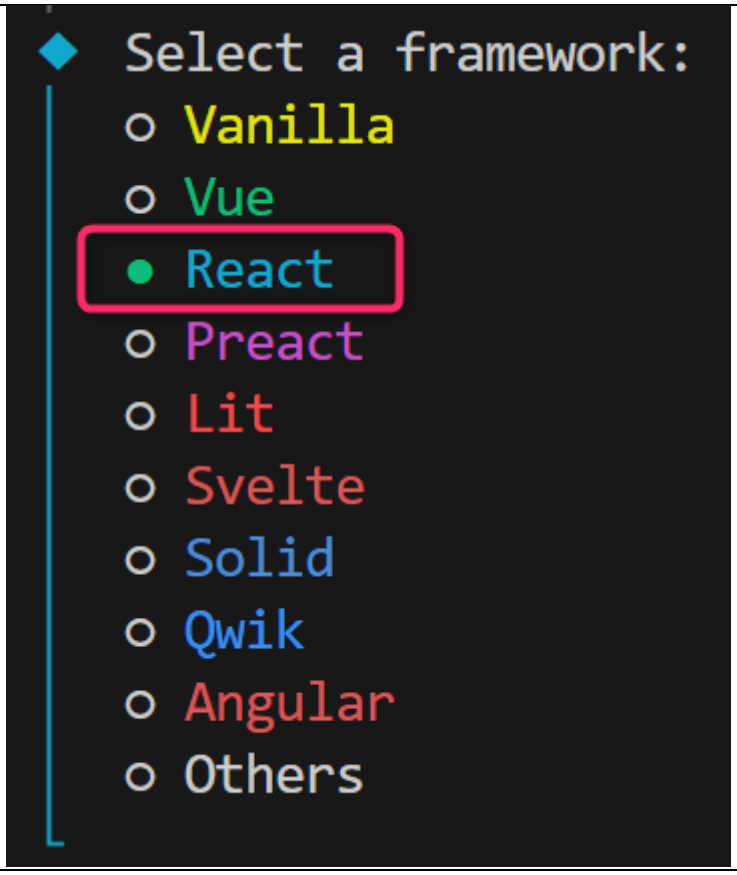
Vite (French for "fast") is a modern build tool and development server that significantly improves the React development workflow. Unlike CRA, which relies on Webpack for bundling, Vite uses esbuild for lightning-fast dependency pre-bundling and native ESM (ECMAScript Modules) in the browser, leading to near-instant cold starts and hot module replacement (HMR).

Key Differences: CRA vs. Vite

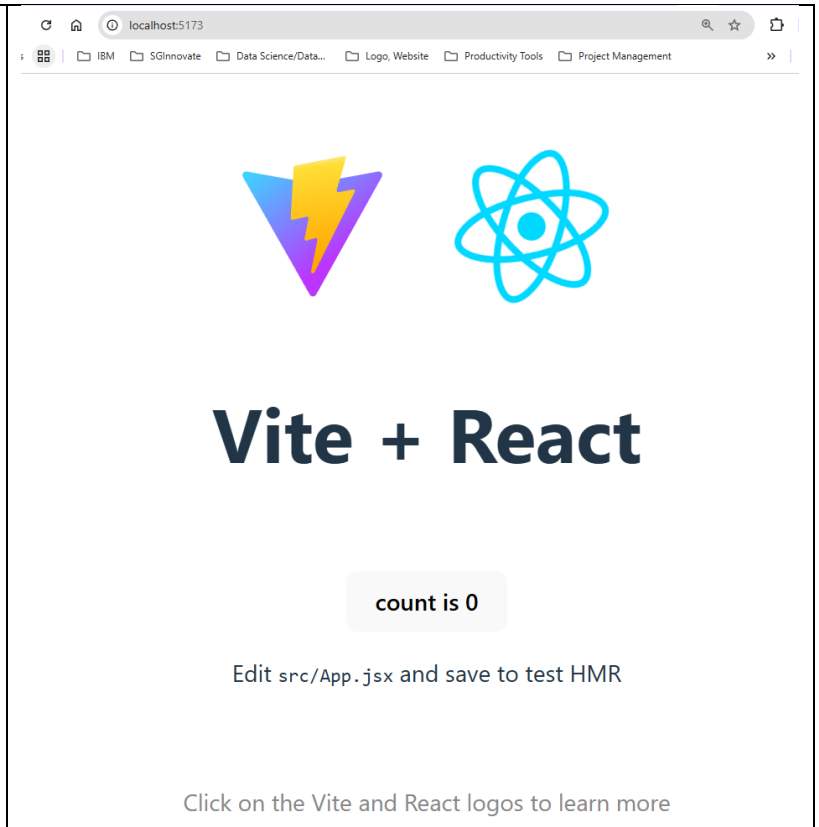
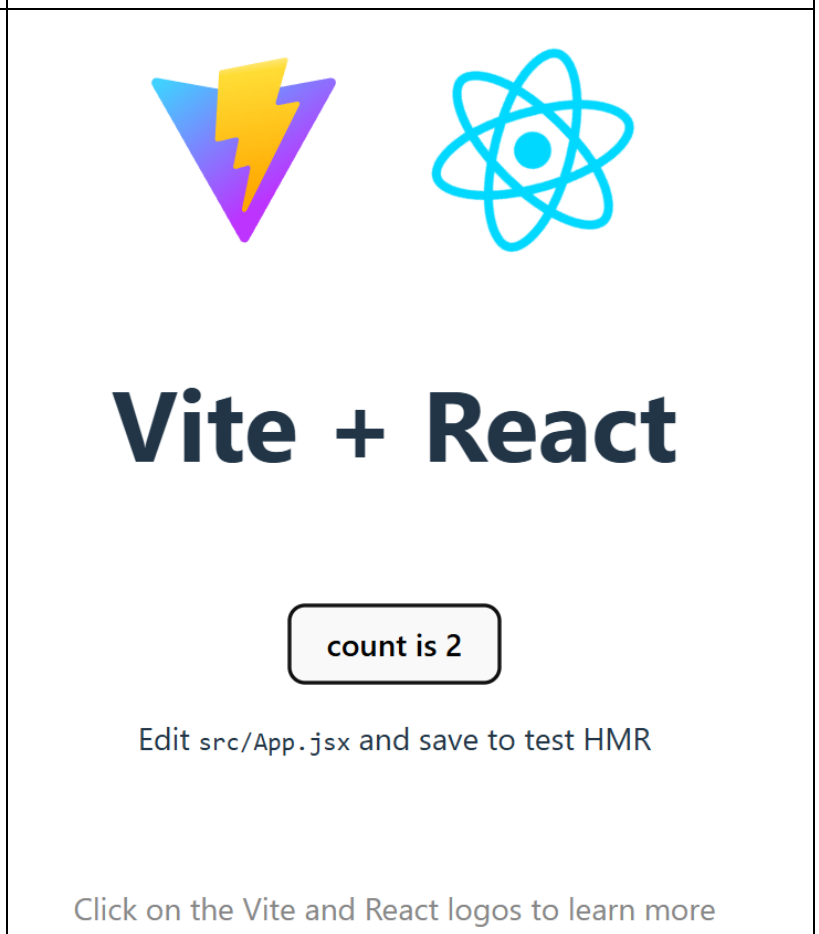
Feature	Create React App (CRA)	Vite
Build Tool	Webpack	esbuild + Rollup
Startup Speed	Slower due to bundling	Instant startup
Hot Module Replacement (HMR)	Slower updates	Fast and efficient
Configuration	Pre-configured Webpack, harder to customize	Lightweight, flexible, easy to extend
Development Server	Node-based	Native ESM, faster refresh
Production Build	Uses Webpack	Uses Rollup for optimized builds
Ecosystem Support	No longer maintained (as of 14 Feb 2025)	Actively developed and widely adopted

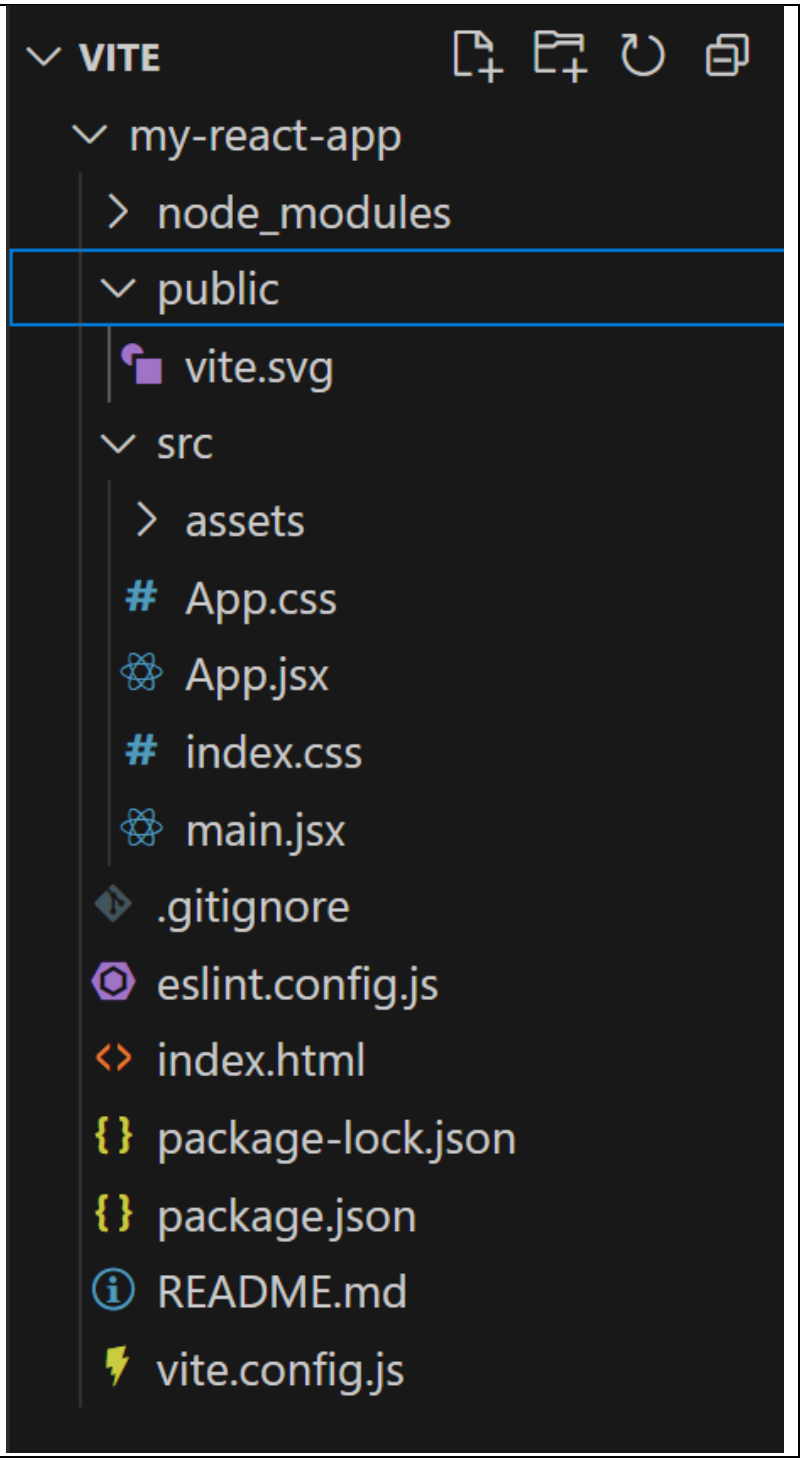
How to create a React App with Vite

In your Visual Studio Code terminal, use **npm create vite@latest project-name –template** to create a new project.
 	
For the first time, it will ask you to install **Vite**.
 	
Type '**y**' to proceed.
	`$ npm create vite@latest my-react-app --template`
	`Need to install the following packages:`
	`create-vite@6.3.1`
	`Ok to proceed? (y) y`

Select the framework for your project: in this case it will be **React**.	

Select the variant: in this case it is **JavaScript**.	◆ Select a variant: ○ TypeScript ○ TypeScript + SWC ● JavaScript ○ JavaScript + SWC ○ React Router v7 ↗
Your project will be created in your folder of choice. Run: **cd my-react-app** to go to the project directory. Run: **npm install** to install all the node modules.	◇ Scaffolding project in [redacted]\vite\my-react-app... Done. Now run: cd my-react-app npm install npm run dev
Run: **npm run dev** to run the my-react-app project. When all is running correctly, you should see the following message.	$ npm run dev > my-react-app@0.0.0 dev > vite VITE v6.2.1 ready in 583 ms → Local: http://localhost:5173/ → Network: use --host to expose → press h + enter to show help

In your http://localhost:5173/ you should see the initial page for the Vite + React app.	
When you click on the count, it will increase the count.	

Vite React App file structure.	

Vite React App File Structure

node-modules folder: contains dependencies installed by npm or yarn. Same as in Create React App (CRA).

public folder: contains static assets. Same as in Create React App (CRA).

src folder: contains the main source code. Same as in Create React App (CRA).

- assets folder: stores images, icons, anything you need to access in your components
- App.css: component styles
- App.jsx: main app component. In CRA, the file is called App.js
- index.css: Global styles
- main.jsx: Root entry file. In CRA, the file is called index.js

gitignore: Specifies untracked files for Git. Same as in Create React App (CRA).

eslint.config.js: ESLint configuration. Same as in Create React App (CRA).

index.html: Main HTML file (entry point). In CRA, this is in the public folder.

package-lock.json: Lock file for package versions. Same as in Create React App (CRA).

package.json: Project metadata & dependencies. Same as in Create React App (CRA).

README.md: Project documentation. Same as in Create React App (CRA).

vite.config.js: Vite configuration file